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Jihwan Choi

MLOps & Software Engineer · SVEvalFlow Owner

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₩448M

SAVED VS AWS

6mo→1.5mo

VALIDATION TIME

99.97%

SUCCESS RATE

80M/day

LOG EVENTS

01 Profile

A software engineer who designs and operates an MLOps inference & evaluation platform for autonomous-driving AI. I co-design the shared Kubernetes GPU cluster (leading scheduler & queue policy) and hold sole ownership of the inference/evaluation workflow (SVEvalFlow: PRISM, Inferit, MTBF Validation).

I cut a 20,000-hour-scale validation that previously took 6+ months on a single machine down to 1.5 months via distributed/parallel processing (173,101 clips, 99.97% success rate), and led a cost-driven On-Premise transition that saved ₩448M (55%) vs AWS. I built a log-analysis system processing 80M daily events that prevented ~₩100M/month in overbilling — consistently turning technical solutions into measurable business value. Most recently I build production Generative AI — Gemini multimodal inference on Vertex AI (running in real customer releases), LLM cost/usage observability at the token- and cost-per-request level, RAG, and agentic tooling — and evangelize AI-native development across the organization.

02 Education

Handong Global University

Feb 2018 — Feb 2022

B.S. — Electronic Engineering, Computer Science

▶ GPA: 3.3 / 4.5

▶ **Coursework:** Data Structures, Algorithms, Computer Architecture, Deep Learning, Machine Learning

03 Experience

Stradvision

Software Engineer
Seoul · Jul 2022 — Present

MLOps Platform — Shared GPU Cluster Co-Design & SVEvalFlow (Inference/Eval) Owner (Oct 2025 — Present)

- ▶ Co-designed and built a shared GPU cluster on RKE2 Kubernetes (60+ heterogeneous GPUs: RTX 3090/4080/4090/5090, RTX 6000); **led scheduler & queue policy design**, integrated 15+ MLOps components (Ray/KubeRay, MLflow, Airflow, ArgoCD, KServe, Harbor).
- ▶ Collaborate on the training workflow (SVDeepFlow); hold **sole ownership of the inference/evaluation workflow (SVEvalFlow)** — Inferit, MTBF, and PRISM below.
- ▶ Built 3 CI/CD pipelines (GitHub Actions + ArgoCD) and Scale-to-Zero serverless inference (KServe/Knative).

Tech: RKE2, Ray, KubeRay, KServe, Knative, MLflow, Airflow, ArgoCD, Harbor, lakeFS, GitHub Actions, Helm

Inferit — svnet3 Build/Inference Acceleration & Artifact Management

- ▶ Built version/config/artifact management for building svnet3 (in-house autonomous-driving vision model) across many configs.
- ▶ Accelerated single-machine inference into distributed/parallel execution across cluster GPUs — cut a 20,000-hour validation from an estimated 6+ months to 1.5 months.
- ▶ Versioned customer option files (e.g., camera parameters), eliminating the 1–2 days of engineer–PM search/communication previously required.
- ▶ 5+ major release versions supported; automated inference per customer issue ticket; integrated Evaluation Driven Development.

Tech: Ray, MLflow, lakeFS, Docker, Kubernetes

MTBF Validation — Large-Scale svnet3 Reliability Validation

- Runs svnet3 inference on customer-ODD driving data before/after release, detecting FP & AEB flags with automated evaluation; processed 173,101 clips at 99.97% success rate, with 205% higher throughput and 67.3% lower processing time vs a single machine.
- Proposed and executed a cost-driven Cloud → On-Premise transition: AWS ₩820M → On-Premise ₩370M (55% saved); ran GPU hardware benchmarks & spec proposals, leveraged HCP Object Storage.
- Automation enabled large-scale validation (previously feasible only at small scale) to be operated by a single person; Visual Inspection UI + **Gemini 2.5 Flash on Vertex AI** auto-FP-detection assist (multimodal, structured output; human decides).

Tech: FastAPI, Airflow, Kubernetes, PostgreSQL, HCP Object Storage, Vertex AI (Gemini)

PRISM — Customer Jira-Integrated ISP Conversion & Re-simulation Pipeline (Nov 2025 →)

- **Phase 1 (deployed/operating):** Event-driven ISP conversion via Jira Webhook → SQS → Lambda; in-house ISP conversion tool integration; ~150 requests/month, 80% faster than manual.
- **Phase 2 (Resim, in dev):** Did the initial development, now technical lead supporting/managing a developer from another org; dual-version inference (issue + latest), Airflow DAG trigger.

Tech: FastAPI, PostgreSQL, AWS (SQS, Lambda), Terraform, Airflow

Data & Labeling Automation (2022 – 2025)

- **WLS:** Analyzed 80M daily labeling-log events to prevent ~₩100M/month overbilling (settlement-audit confirmed); pinpointed bottlenecks feeding UI/UX and the ALAS model (+20% labeling efficiency). (Django, MongoDB, AWS SQS)
- **ALAS:** Auto-labeling microservice, reduced manual annotation workload by ~35%. (FastAPI, AWS ECS/EKS, CDK)
- **IGS:** External-customer data ingestion service planning & architecture design (design-only). (Django, Airflow, Kubernetes)
- **SURF Compass:** Device monitoring service planning & design (design-only). (FastAPI, PostgreSQL)
- **RNS:** Automated release-note generation, 50% faster authoring (4h → 2h). (NestJS, Atlassian API)
- **SLT:** 2D LD labeling tool, improved productivity by 30%. (C++/MFC)

Korea Institute of Robotics & Technology Convergence

Researcher
Pohang · Jan – Jul 2022

- Developed a real-time gesture-based Human-Robot Interaction system.
- Supported an autism-diagnosis assistance robot project.

04 Skills

Languages	Python · C++ · TypeScript
Backend	Django · FastAPI · NestJS
Database	PostgreSQL · MongoDB · Redis
Cloud	GCP (Vertex AI) · AWS (EC2, EKS, ECS, SQS, SNS, Lambda, S3, Bedrock)
Infrastructure	Kubernetes (RKE2) · Docker · Terraform · CDK · Rancher
ML / ML0ps	Ray · KubeRay · MLflow · KServe · Knative · lakeFS
Workflow / CI-CD	Airflow · Celery · ArgoCD · GitHub Actions · Harbor · Helm
Monitoring	Grafana · Prometheus · Loki · DataDog
Generative AI / LLM	Vertex AI (Gemini) · AWS Bedrock · multimodal · structured output · RAG / embeddings · prompt engineering · LLM cost & observability · agentic / tool-use · multi-model orchestration · Claude Code / MCP / Skills
Tools	Git · Jira · Confluence · Atlassian API

05 Generative AI Engineering & Internal Evangelism

- ▶ **Gemini on Vertex AI (production, multimodal):** Gemini 2.5 Flash service on Vertex AI for auto-FP detection on AD vision inference (image + structured metadata); JSON-schema structured output, per-request token tracking, batched async with deduplication (~40% fewer calls), human-in-the-loop; runs in real customer-release evaluation (MTBF).
- ▶ **LLM cost & usage observability (AWS Bedrock):** Token-level and per-model/per-team cost-per-request attribution via CloudWatch + Slack with Terraform IaC; 90%+ accuracy vs billing, adopted as an org cost-optimization tool.
- ▶ **RAG knowledge system (LLM Wiki):** Hybrid keyword + embedding retrieval (text-embedding-3-small, vector index, cosine top-k) → grounded answers with citations and freshness; CI-rebuilt index (Git + Markdown).
- ▶ **Agentic LLM tooling:** NL intent → cluster diagnostics/storage tools (Claude Code Skills) with multi-step orchestration and guardrails (dry-run, rollback, gates); multi-model self-critique workflow across Claude/GPT/Gemini (github.com/jhwanchoi/dotfiles).
- ▶ Hosted internal AI-driven development meetings; evangelized modern AI dev tools (Claude Code, Gemini CLI, Codex; Plugin/MCP/Skill); n8n automation prototypes.

Tech: Vertex AI (Gemini), AWS Bedrock, embeddings/RAG, Claude Code, Gemini CLI, Codex, MCP, CloudWatch, Terraform, n8n